

Course 3

FUNDAMENTALS of MACHINE LEARNING for HEALTH CARE

Module 1

Healthcare ecosystem

- AI developer
- tech company
- policy maker
- regulator
- health care system leadership
- pharmas
- medical device
- frontline clinicians
- ethicist
- patients
- carers

standard unit
med experimental computer

as med

SUMEX - AIM

reproduction

major (net), DG, to
AI handbook projects

1970s - New England Journal of Med → Computer Science displacing workforce

AI needed

- AI algorithms for complex models
- access to large volumes of data
- powerful graphical processing unit or GPU's → massive parallel computation

AI can help healthcare systems

- improve access
- workforce balance
- ↑ quality

Unresolved Concerns about AI in healthcare

- Workforce displacement
- Skill atrophy
- algorithmic + over bias
- Patient privacy
- medico-legal responsibility
- oversight + regulation

early application of ML in healthcare

- care delivery (enhancing, optimising)
- info MGMT (integrating)
- cognitive support for clinical care
- prediction
- risk assessment
- early detection

Machine learning

- used by data scientist
- statistical

→ statistical + mathematical modelling techniques that learn from data without explicit programming

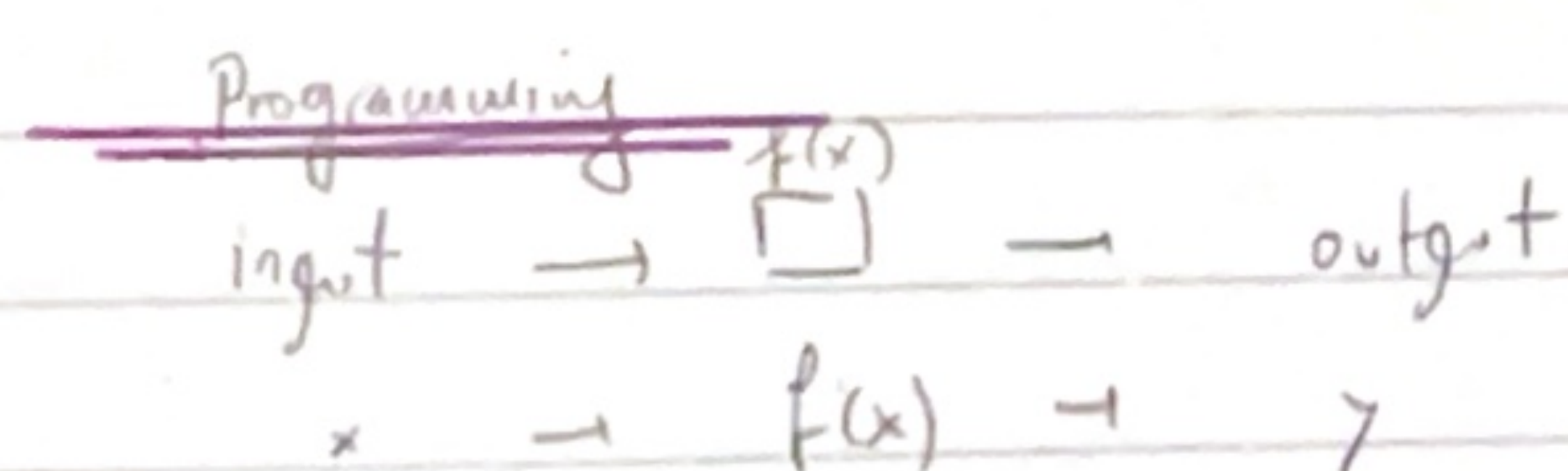
algebra	} math foundations
statistics	
probability	

AI

- used for dissemination

- development of the development of machine capabilities to simulate learning or other features of intelligence -

	<u>Statistics</u>	vs.	<u>ML</u>
bl	statistics, data source		computer science, engineering
approach	hypothesis-driven model development		creating system that learn from data
goal	inferences relationships between variables		optimization prediction accuracy
assumptions	need population info		—
success	discovers that can be applied for new predictions learn new insights		resulting model produces accurate predictions without predefined characteristics ↓ more performance based



ML

Code all $\neq$ permutations and
decisions with the input data
for an accurate output

Just by leveraging existing data,
ML models learn all components of
a function (that would have had to
be coded by someone)

↓

learn to assign good values and
calculations based on data

leverage $\rightarrow$ aggregator